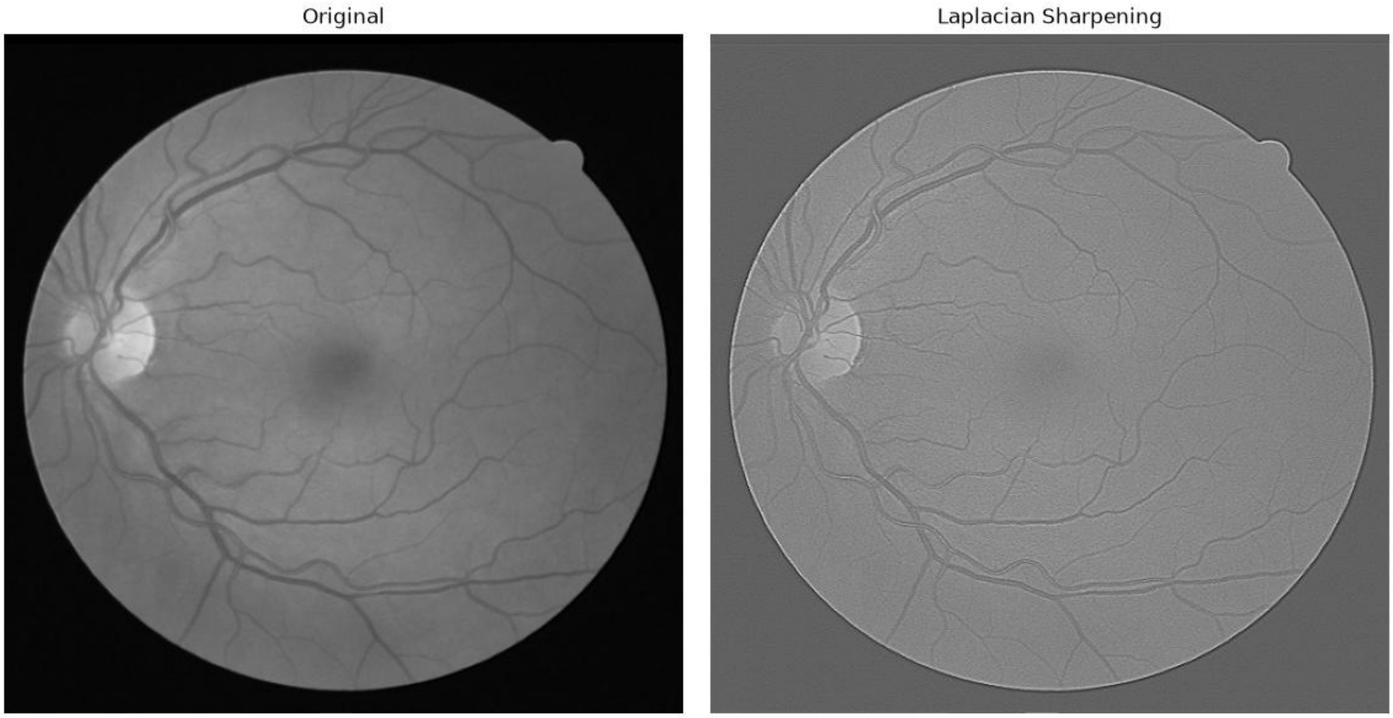
```python
# Code for sharpening one image using Laplacian Sharpening (Roll: 2107042)

image = cv2.imread(r"Inputs/homework_a2_b2.png", cv2.IMREAD_GRAYSCALE)
c = 3
# Laplacian Kernel
kernel = np.array([
    [0, -1, 0],
    [-1, 4, -1],
    [0, -1, 0]
], dtype=np.float32)

kernel = np.flip(kernel)

def manual_convolution(image, kernel):
    h, w = image.shape
    kh, kw = kernel.shape
    pad = kh // 2
    padded = np.zeros((h + 2 * pad, w + 2 * pad), dtype=image.dtype)

    for i in range(h):
        for j in range(w):
            padded[i + pad, j + pad] = image[i, j]

    output = np.zeros((h, w), dtype=np.float32)

    for i in range(h):
        for j in range(w):
            sum = 0
            for m in range(kh):
                for n in range(kw):
                    sum += padded[i + m, j + n] * kernel[m, n]
            output[i, j] = sum

    return output

# Choose One
laplacian = manual_convolution(image, kernel)
# laplacian = cv2.filter2D(image, cv2.CV_32F, kernel)

# Laplacian Sharpening
output = image.astype(np.float32) + c * laplacian

output = cv2.normalize(
    output, None, 0, 255,
    cv2.NORM_MINMAX
).astype(np.uint8)
```


Original
Laplacian Sharpening